Noise Beats Coarseness: The Value of Inside Information in the Kyle-Back Model

A coarse signal earns its explained variance, a noisy normal signal its square root

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Abstract

Two signals that explain the same share of a normal fundamental's variance can be worth very different amounts to an insider in the continuous-time Kyle model. The sign of the fundamental explains 63.7 percent of its variance and earns 63.7 percent of the full-information profit, while normal noisy information of the same precision earns 79.8 percent. In general the insider's profit share lies between R^2 and R, where R^2 is the share of fundamental variance his information explains. Normal noise attains the upper bound and information that sorts the fundamental into ordered categories attains the lower bound exactly. Among signals with n messages the best is the Lloyd-Max quantizer of the normal law, which earns 63.7, 81.0 and 88.3 percent for $n = 2, 3, 4$. The gap arises because order flow must end at a normally distributed point for the insider to stay hidden. Inside the model the covariation of price changes with order flow equals the value of information for every signal, while the ratio of fundamental variance to price impact, a proposed approximation, overstates the value of sign information by the factor $(\pi/2)^2 = 2.47$. Replacing fundamental variance with price variance still leaves a factor of $\pi/2$. Non-normal information also reaches prices more slowly.

1 Introduction

How much is inside information worth when it is coarse? Much private information comes in categories: a deal will or will not close, earnings will beat or miss, a rating will move up, down or not at all. In the Kyle-Back model with a normal fundamental, information of this kind earns the insider exactly the share R^2 of the full-information profit, where R^2 is the share of fundamental variance it explains. Normal information with the same R^2 earns R, the square root, which is always more (Theorem 1). Every other signal earns something in between. The sign of the fundamental, for instance, explains $2/\pi = 63.7$ percent of its variance and earns exactly that share, while normal noise of the same precision earns 79.8 percent.

The reason is that the insider must hide in noise. In Back's (1992) equilibrium the insider steers total order flow to an endpoint that is normal, as noise trading is, and his profit is the covariance between his valuation and that endpoint. A normal valuation can be matched one for one to a normal endpoint, while the best match for a two-point valuation has correlation $\sqrt{2/\pi} = 0.798$. The value of information therefore depends on its shape as well as its precision.

The same logic answers a design question. If the insider can receive at most n distinct messages, the most valuable signal tells him which cell of the Lloyd-Max quantizer of the normal law contains the fundamental, and it earns one minus the quantizer's distortion (Theorem 2). Three categories earn 81.0 percent of perfect foresight and four earn 88.3 percent. Randomizing or adding noise does not help.

The shape of information also decides which measure of its value is accurate. Kadan and Manela (2026) define the value of information as the covariation of price changes with order flow, and inside this model that covariation equals the insider's profit for every information structure (Corollary 1). Kadan and Manela (2025) instead use the ratio of fundamental variance to price impact, which approximates the value of information in the model of Back, Cao and Willard (2000). With one insider the ratio equals the value of information divided by $(R\rho)^2$, where $\rho \leq 1$ measures how close the insider's valuation is to normal in shape, so imprecision and coarseness both inflate it. For sign information the ratio overstates the value by $(\pi/2)^2 = 2.47$. Using price variance in place of fundamental variance removes the precision factor and leaves the shape factor, $\pi/2$ for sign information. For any monotone coarse signal the price-variance ratio equals $\sigma_u\sigma_v$, the full-information profit, so it cannot tell an insider who knows only the sign of the fundamental from one who knows the fundamental itself.

Non-normal information also reaches prices more slowly. By date t prices reflect at most the share t of the insider's information, with equality only for normal information, and for sign information the share is $(2/\pi)\arcsin t$, one third at $t = 1/2$ (Proposition 1).

A reader may object that coarse information is a special case and that the model is stylized. Theorem 1 applies to every signal, and categorical information is simply the case in which its lower bound binds. The second objection has more force, since the results use one risk-neutral insider, competitive market makers who know the information structure, continuous trading and a normal fundamental, and with several insiders or discrete trading the bounds need not hold. The comparison of measures is an identity inside a one-insider model and says nothing about the size of any bias in Kadan and Manela's estimates.

Several building blocks are known. The profit representation is implicit in Back (1992) and explicit in transport form in Back, Cocquemas, Ekren and Lioui (2021). The upper bound R follows from Gelbrich's (1990) bound, which Viswanathan and Xing (2026) use to show that, for a normal prior and under a regularity condition, a normal signal is optimal when information costs are proportional to mutual information. We have not found the lower bound, its attainment by coarse signals, Theorem 2, the factors in Corollary 1 or the speed bound in the work we searched.

2 Setting

The fundamental v has mean zero and variance σ_v^2 and is paid at date 1. Noise traders' cumulative orders form a Brownian motion U with volatility σ_u. A risk-neutral insider observes a signal s at date 0 and submits orders X. Competitive risk-neutral market makers observe total order flow $Y = X + U$ and set $P_t = \mathbb{E}[v \mid Y_s,\ s \le t]$. The joint law of (v, s) is common knowledge. The insider's valuation is $m = \mathbb{E}[v \mid s]$, with distribution function F, and the model is Back's model with liquidation value m.

In equilibrium (Back 1992), let $h(y) = F^{-1}(\Phi(y/\sigma_u))$ map a $N(0, \sigma_u^2)$ variable into the law of m, and let $P(t, y) = \mathbb{E}[h(y + \sigma_u\sqrt{1-t}\,Z)]$ with $Z \sim N(0,1)$. The insider trades so that Y is a Brownian bridge ending at a point ξ with $h(\xi) = m$; when F has atoms he draws ξ from $N(0, \sigma_u^2)$ restricted to the interval that h maps to m. To market makers Y is a Brownian motion, so $Y_1 \sim N(0, \sigma_u^2)$ and $P_1 = m$. Back (1992) treats continuous F, and the construction extends to atoms. Let $\varphi' = h$ and $\Phi(t, y) = \mathbb{E}[\varphi(y + \sigma_u\sqrt{1-t}\,Z)]$, so that $P = \partial_y\Phi$. Itô's formula shows that every absolutely continuous strategy earns $\mathbb{E}[mY_1 - \varphi(Y_1)]$ plus a constant. Because h is nondecreasing, φ is convex, and any endpoint with $h(Y_1) = m$ maximizes $my - \varphi(y)$; the randomized endpoints keep $Y_1 \sim N(0, \sigma_u^2)$.

Lemma 1 (Profit)**.** *The insider's expected profit is*

$$\Pi = \mathbb{E}[mY_1] = \sigma_u \int_0^1 F^{-1}(p)\,\Phi^{-1}(p)\,dp = \sigma_u \max_{\xi \sim N(0,1)} \mathbb{E}[m\,\xi],$$

where the maximum is over couplings of m with a standard normal variable.

Proof. Profit is $\mathbb{E}\int_0^1 (v - P_t)\,dX_t$ with $X = Y - U$. Because X is adapted to the insider's information, v can be replaced by m. The term $\mathbb{E}\int (m - P_t)\,dU_t$ is zero because U is a Brownian motion for the insider, and $\mathbb{E}\int P_t\,dY_t$ is zero because Y is a Brownian motion and P a martingale for market makers. What remains is $\mathbb{E}[mY_1]$. In equilibrium Y_1 is the increasing rearrangement of m onto the normal law, which attains the largest covariance among all couplings (Hardy-Littlewood). □

With full information $m = v$ and $\Pi_{\text{full}} = \sigma_u\sigma_v$. We call $\Pi/(\sigma_u\sigma_v)$ the profit share and write $R^2 = \mathrm{Var}(m)/\sigma_v^2$. By Stein's lemma, expected price impact $\bar{\lambda} = \mathbb{E}[\partial_y P(t, Y_t)] = \mathbb{E}[h'(Y_1)] = \Pi/\sigma_u^2$ at every date.

3 Coarse information earns its explained variance

Theorem 1. *Let $v \sim N(0, \sigma_v^2)$ and let s be any signal. Then*

$$R^2 \;\le\; \frac{\Pi}{\sigma_u\sigma_v} \;\le\; R.$$

The upper bound holds with equality if and only if m is normal. The lower bound holds with equality if and only if m is almost surely a nondecreasing function of v.

Proof. By Lemma 1, Π/σ_u is the largest value of $\mathbb{E}[m\xi]$ over couplings of m with a standard normal ξ. Cauchy-Schwarz gives $\mathbb{E}[m\xi] \le \mathrm{sd}(m) = R\sigma_v$, with equality only if m is proportional to ξ and hence normal.

For the lower bound, v/σ_v is itself a standard normal coupled with m, and $\mathbb{E}[mv]/\sigma_v = \mathbb{E}[m^2]/\sigma_v = R^2\sigma_v$ by iterated expectations. For equality, Hoeffding's identity writes the covariance of a coupling as $\iint [H(x,y) - F(x)G(y)]\,dx\,dy$, where H is the joint distribution function, and $H \le \min(F,G)$. The covariance is therefore maximal only for the comonotone coupling, whose law is unique. The coupling (m, v) attains it if and only if m and v are comonotone, and since v has no atoms this means m is a nondecreasing function of v. □

Monotone coarse information includes any signal of the form "v lies in the i-th of n consecutive intervals", thresholds, and censored signals such as "v itself if $|v| > c$, otherwise nothing". Its value is one minus the mean squared error of the cell means, in units of σ_v^2. The sign of v gives $m = \pm\sigma_v\sqrt{2/\pi}$ and a share of $2/\pi = 0.637$. A signal whose valuation is identically zero, such as $\mathbf{1}\{|v| > a\}$, earns nothing.

At given explained variance the insider prefers noise to coarseness. At $R^2 = 0.25$, normal noise earns half of perfect foresight and an ordered partition earns a quarter. Other signals lie strictly between the bounds. The sign of v observed with noise, at $R^2 = 0.318$, earns 0.450; the non-monotone signal $\mathbf{1}\{0.5 < v/\sigma_v < 1.5\}$ has $R^2 = 0.270$ and earns 0.379.

In general the share equals $R\rho$, where $\rho = \max \mathbb{E}[m\xi]/\mathrm{sd}(m)$ is the largest correlation between m and a normal variable. Equivalently $\rho = 1 - \frac{1}{2}W_2^2$, with W_2 the Wasserstein distance between the standardized law of m and the standard normal. Theorem 1 says $\rho = R$ for monotone coarse information. Table 1 gives ρ for some laws of m.

Law of m	Two-point	Uniform	Laplace	Student t_5	Student t_3
ρ	0.798	0.977	0.981	0.983	0.910
$1/\rho^2$	1.571	1.047	1.038	1.034	1.208

Table 1: Largest correlation ρ between the insider's valuation and a normal variable, for symmetric laws. The two-point value is $\sqrt{2/\pi}$.

4 The best signal with n messages

Theorem 2. *Let $v \sim N(0, \sigma_v^2)$. If $\mathbb{E}[v \mid s]$ takes at most n values, then*

$$\frac{\Pi}{\sigma_u\sigma_v} \;\le\; 1 - D_n, \qquad D_n = \min_q \mathbb{E}\big[(Z - q(Z))^2\big], \quad Z \sim N(0,1),$$

where the minimum is over quantizers q with n levels. Equality holds for the signal "which Lloyd-Max cell contains v/σ_v" (Lloyd 1982; Max 1960).

Proof. Let m take values $m_1 < \cdots < m_k$, $k \le n$, with probabilities p_i. Let $I_1, \ldots, I_k$ be consecutive intervals with normal probabilities p_i, and $\bar{z}_i = \mathbb{E}[Z \mid Z \in I_i]$. Lemma 1 gives $\Pi/\sigma_u = \sum_i p_i m_i \bar{z}_i$. Put $T_j = \sum_{i>j} p_i m_i = \mathbb{E}[v\,\mathbf{1}\{m > m_j\}]$. Among events of a given probability, $\mathbb{E}[v\mathbf{1}_A]$ is largest for an upper tail of v, so $T_j \le \sigma_v U_j$ with $U_j = \sum_{i>j} p_i \bar{z}_i$. Because $\mathbb{E}[m] = 0$ and the $\bar{z}_i$ increase, summation by parts gives

$$\sum_i p_i m_i \bar{z}_i = \sum_{j=1}^{k-1} (\bar{z}_{j+1} - \bar{z}_j)\, T_j \;\le\; \sigma_v \sum_{j=1}^{k-1} (\bar{z}_{j+1} - \bar{z}_j)\, U_j = \sigma_v \sum_i p_i \bar{z}_i^2.$$

The last sum is one minus the mean squared error of the quantizer with cells I_i and centroid levels, which is at least $D_k \ge D_n$. The Lloyd-Max cell signal makes m monotone in v, so Theorem 1 gives it the share $R^2 = 1 - D_n$. □

Since only m enters, noisy or randomized signals with n messages do no better. Table 2 reports the values. The optimal cells are the sign for $n = 2$, thresholds at $\pm 0.612\,\sigma_v$ for $n = 3$, and thresholds at 0 and $\pm 0.982\,\sigma_v$ for $n = 4$. For large n the Panter-Dite (1951) approximation $D_n \approx 2.72/n^2$ applies, so a signal with n categories gives up about $2.72/n^2$ of the profit, while a normal signal carrying $\log_2 n$ bits gives up about $1/(2n^2)$.

The last column uses a bound in bits. If $I(v; s) \le k \ln 2$, Shannon's lower bound for a normal source gives $\mathbb{E}[(v-m)^2] \ge \sigma_v^2 4^{-k}$, so $R^2 \le 1 - 4^{-k}$ and, by Theorem 1, the share is at most $\sqrt{1-4^{-k}}$, attained by normal noise. One bit is worth at most $\sqrt{3}/2 = 86.6$ percent of perfect foresight. That normal signals are optimal under a mutual-information cost is the result of Viswanathan and Xing (2026); the bound restates it as a capacity constraint. The comparison with Table 2 mixes shape and precision, since at one bit the sign has $R^2 = 0.637$ and the normal signal 0.75. The clean comparison of shape holds R^2 fixed, as in Theorem 1.

Messages n	Profit share $1-D_n$	Entropy (bits)	Normal signal with the same bits
2	0.6366	1.000	0.8660
3	0.8098	1.536	0.9386
4	0.8825	1.911	0.9640
5	0.9201	2.203	0.9761
8	0.9655	2.825	0.9900

Table 2: Best profit share with n messages, the entropy of the optimal signal, and the share earned by a normal signal carrying the same number of bits, $\sqrt{1-4^{-k}}$ for k bits.

5 Two consequences

Corollary 1. *Let m have any law with finite variance and let $\rho = \Pi/(\sigma_u \operatorname{sd}(m))$. Then*

$$\mathbb{E}\int_0^1 dP_t\, dY_t = \bar{\lambda}\sigma_u^2 = \Pi, \qquad \frac{\operatorname{Var}(P_1 - P_0)}{\bar{\lambda}} = \frac{\Pi}{\rho^2}, \qquad \frac{\sigma_v^2}{\bar{\lambda}} = \frac{\Pi}{(R\rho)^2}.$$

Proof. The first identity is the price-impact fact after Lemma 1. For the second, $\operatorname{Var}(P_1 - P_0) = \operatorname{Var}(m)$ and $\bar{\lambda} = \Pi/\sigma_u^2 = \rho \operatorname{sd}(m)/\sigma_u$. For the third, $\operatorname{sd}(m) = R\sigma_v$, so $\Pi = R\rho\,\sigma_u\sigma_v$ and $\sigma_v^2/\bar{\lambda} = \sigma_u^2\sigma_v^2/\Pi = \Pi/(R\rho)^2$. □

The three measures differ by two factors. The covariation of price changes with order flow, the measure of Kadan and Manela (2026), equals the value of information for every information structure. The ratio of price variance to price impact divides it by ρ^2 and is exact only when m is normal (Table 1). The ratio of fundamental variance to price impact, the measure of Kadan and Manela (2025), divides it by $(R\rho)^2$ and adds the precision factor R^2. Table 3 reports both factors for information about a normal fundamental. For monotone coarse information $\rho = R$, so the price-variance ratio equals $\sigma_u\sigma_v$, the full-information profit, whatever the coarseness, and the fundamental-variance ratio equals $\sigma_u\sigma_v/R^2$. The quotient of the covariation and the price-variance ratio identifies ρ^2, a measure of how far the insider's information is from normal. Kadan and Manela (2025) derive their ratio as an approximation in the model of Back, Cao and Willard (2000) with several insiders, and the comparison here is an identity inside a one-insider model, so we do not apply it to their estimates.

Information about $v \sim N(0,\sigma_v^2)$	Covariation	Price var./impact	Fund. var./impact
Normal noise, explained share R^2	1	1	$1/R^2$
Monotone coarse, explained share R^2	1	$1/R^2$	$1/R^4$
Sign of v ($R^2 = 2/\pi$)	1	1.571	2.467
Three Lloyd-Max categories ($R^2 = 0.810$)	1	1.235	1.525

Table 3: Each measure divided by the value of information Π, from Corollary 1. For normal noise $\rho = 1$; for monotone coarse information $\rho = R$.

Proposition 1. *Let m have any law with finite variance and expand $h(\sigma_u z) = \sum_k c_k\, He_k(z)/\sqrt{k!}$ in Hermite polynomials. Then $\bar{\lambda} = c_1/\sigma_u$ and*

$$\operatorname{Var}(P_t) = \sum_{k\geq 1} c_k^2\, t^k, \qquad \frac{\operatorname{Var}(P_t)}{\operatorname{Var}(m)} \leq t,$$

with equality at an interior date if and only if m is normal. For sign information $\operatorname{Var}(P_t)/\operatorname{Var}(m) = (2/\pi)\arcsin t$.

Proof. $P_t = \mathbb{E}[h(Y_1) \mid Y_t]$, and (Y_t, Y_1) is bivariate normal with correlation $\sqrt{t}$. Mehler's formula gives $\mathbb{E}[He_k(Y_1/\sigma_u) \mid Y_t] = t^{k/2} He_k\big(Y_t/(\sigma_u\sqrt{t})\big)$, and orthogonality gives the series. Since $t^k \leq t$ for $k \geq 1$, the bound follows, with equality at $t \in (0,1)$ only if $c_k = 0$ for $k \geq 2$. For sign information $\operatorname{Var}(P_t)$ is the covariance of $\operatorname{sign}(Y_1)$ and $\operatorname{sign}(Y_1')$, two endpoints drawn independently given Y_t, whose correlation is t; the arcsine law gives $(2/\pi)\arcsin t$. □

Sign information is one third impounded at $t = 1/2$, against one half for normal information. Three Lloyd-Max categories reach 0.20, 0.41 and 0.63 at $t = 0.25$, 0.5 and 0.75, and five reach 0.23, 0.46 and 0.70. Holding $R^2 = 2/\pi$ fixed, sign information produces expected price impact $0.637\,\sigma_v/\sigma_u$ and normal information $0.798\,\sigma_v/\sigma_u$, 25 percent more. The bound places normal information above every other shape and leaves the ranking among non-normal signals open.

6 Scope

All results hold in a stylized model with one risk-neutral insider, continuous trading over a fixed horizon and competitive market makers who know the information structure but not the signal. Theorems 1 and 2 need a normal fundamental; for other priors Lemma 1 still prices any signal, but R^2 no longer summarizes it. Nothing here is tested on data. The numerical values were computed by quadrature, bivariate normal probabilities and iteration of the Lloyd-Max conditions.

References

[1] Back, K., F. Cocquemas, I. Ekren and A. Lioui (2021). Optimal transport and risk aversion in Kyle's model of informed trading. arXiv:2006.09518.

[2] Back, K., C. H. Cao and G. A. Willard (2000). Imperfect competition among informed traders. *Journal of Finance* 55(5), 2117–2155.

[3] Back, K. (1992). Insider trading in continuous time. *Review of Financial Studies* 5(3), 387–409.

[4] Gelbrich, M. (1990). On a formula for the L^2 Wasserstein metric between measures on Euclidean and Hilbert spaces. *Mathematische Nachrichten* 147, 185–203.

[5] Kadan, O., and A. Manela (2025). Liquidity and the strategic value of information. *Review of Finance* 29(1), 1–32.

[6] Kadan, O., and A. Manela (2026). The value of information: a puzzle. arXiv:2605.11180.

[7] Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53(6), 1315–1335.

[8] Lloyd, S. P. (1982). Least squares quantization in PCM. *IEEE Transactions on Information Theory* 28(2), 129–137.

[9] Max, J. (1960). Quantizing for minimum distortion. *IRE Transactions on Information Theory* 6(1), 7–12.

[10] Panter, P. F., and W. Dite (1951). Quantization distortion in pulse-count modulation with nonuniform spacing of levels. *Proceedings of the IRE* 39(1), 44–48.

[11] Viswanathan, S., and H. Xing (2026). Flexible information acquisition in the Kyle model. arXiv:2603.21842.